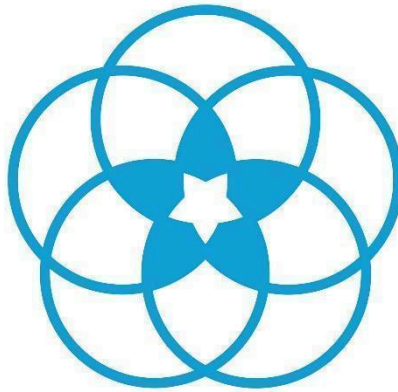




Health Information System Program Middle East and North Africa (HISP MENA)

Integration Documentation: Hakeem
Integration API Documentation

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HISP MENA

Table 1: Version control

Version	Date	Description of Changes
1.0	2025-09-05	Initial creation of the document.
1.1	2025-09-15	Added detailed examples for dry-run vs. commit uploads and clarified scheme parameter behavior.
1.2	2025-09-25	Updated response explanation section with additional import result cases and improved troubleshooting notes.
1.3	2025-11-01	Added the “Postman Quick-Start Kit” section and refined token generation instructions for Hakeem system user.

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1. Purpose

Upload data from Hakeem to DHIS2 via the middleware system. Payloads are agreed to use CSV files for speed and smaller size.

2. Endpoint & Transport

Property	Value
Method	POST
URL (template)	http://<middleware-host>/camel/v1/datasets?transactionUUID=xxx
Protocol	HTTP

3. Authorization

Field	Value
Type	Bearer Token
Header	Authorization: Bearer <token> For testing purposes, you can use the following token: eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiJoYWtlZW1AbWV4cCI6MjA4MzE5MDQwMCwicm9sZXMiOiIiSGFrZWVtIHJvbGUiXSwicGVybnWlzc2lvbnMiOiJldLCJyb3V0ZXMiOiZGhpczItb2FkLWNzdiJdfQ.taSuQ2hohUlwrR7Ic1LLoR-ulzGGKFikoZlkPmoRhu_s

4. How Hakeem Gets the Token

- Send a POST request to:

`http://<middleware-host>/api/auth/system-user-token?days=7`

Parameter	Type	Required	Description
days	integer	yes	Number of days before the token expires. Used to control token lifetime.

Body (JSON):

```
{  
  "email": "hakeem@mail.com",  
  "password": "Ss123@231"  
}
```

- The response will return a token like this:

```
{  
  "token": "eyJhbGciOiJIUzI1NiJ9...."  
}
```

- Use this token in all uploads by adding it to the header:

Authorization: Bearer <token>

5. Inputs

HTTP Headers

Header	Required	Value	Notes
Authorization	Yes	Bearer <token>	Access control
Content-Type	Yes	application/csv	MUST be CSV; body is raw file bytes

Required Query Parameters

Name	Type/values	Required	Description
transactionUUID	UUID v4	Yes	Unique ID for each upload; used for tracking/idempotency.
scheme	UID CODE NAME	No	Identifier scheme for matching in RHIS.
strategy	append update delete	No	Import action applied to incoming rows. Choose per business need.
dry-run	true false	No	Validation mode; true=validate only. Server default is true

Request Body

Type: Binary file upload (one CSV per request). Ensure the client sends non-zero Content-Length.

6. Outputs

HTTP Status Codes

Code	Meaning
200	Accepted/validated successfully (dry run or commit)
400	Bad Request: missing/empty/invalid transactionUUID, or empty CSV body
409	Conflict: duplicate transactionUUID
500	Internal Server Error during parsing/processing

Response Body

Success returns a JSON body including a tracking identifier when available.

Example (200 OK):

```
{
  "status": "SUCCESS",
  "message": "completed successfully",
  "errorCode": "",
  "errorMessage": "",
  "details": {
    "httpStatus": "OK",
    "httpStatusCode": 200,
    "status": "OK",
    "message": "Import was successful.",
    "response": {
      "responseType": "ImportSummary",
      "status": "SUCCESS",
      "importOptions": {
        "idSchemes": {
          "idScheme": {"type": "CODE"},
          "dataElementIdScheme": {"type": "CODE"},
          "categoryOptionComboIdScheme": {"type": "CODE"},
          "orgUnitIdScheme": {"type": "CODE"}
        },
        "dryRun": false,
        "importStrategy": "UPDATES",
        "mergeMode": "REPLACE",
        "reportMode": "FULL",
        "firstRowIsHeader": true
      },
      "description": "Import process completed successfully",
      "importCount": { "imported": 0, "updated": 0, "ignored": 6, "deleted": 0 },
      "conflicts": [],
      "rejectedIndexes": []
    }
  }
}
```

```

    "dataSetComplete": "false"
  }
}
}

```

Explanation:

How to read the above response.

Field	Meaning
status = SUCCESS	The overall process succeeded.
importOptions.idSchemes.* = CODE	All lookups used DHIS2 codes (not UIDs).
importStrategy = UPDATES	Only update existing data; no new rows created.
mergeMode = REPLACE	Incoming values overwrite existing ones on match.
dryRun = false	This run committed changes (not just validation).
importCount.imported = 0	No brand-new rows were created.
importCount.updated = 0	No existing rows were changed.
importCount.ignored = 6	6 rows were skipped (no effective change or no resolvable match).
conflicts = []	The server reported no row-level errors.

Why can "ignored" be > 0 on SUCCESS.

Likely reason	What to check
Rows identical to existing values	If strategy is UPDATES, unchanged values may be ignored.
Code mapping didn't resolve	Verify orgUnit/dataElement/categoryOptionCombo codes exist in RHIS.
Period or dataset rules skipped the row	Confirm dataset/period alignment and required combos.

Header/column mismatch	Ensure firstRowIsHeader=true and columns match expected format.
-------------------------------	---

7. Examples

- Minimal upload (defaults)

POST <https://<host>/camel/v1/datasets?transactionUUID=<uuid>>

- Validate only (dry run)

POST <https://<host>/camel/v1/datasets?transactionUUID=<uuid>&dry-run=true>

- Append without commit

POST <https://<host>/camel/v1/datasets?transactionUUID=<uuid>&strategy=append&dry-run=true>

- Update with commit

POST <https://<host>/camel/v1/datasets?transactionUUID=<uuid>&strategy=update&dry-run=false>

8. Error Scenarios & Fixes (from testing collection - Postman)

Case	Resolution
Missing transactionUUID	Add transactionUUID=<uuid> to the query string.
Empty transactionUUID	Provide a valid UUIDv4 (e.g., Postman {{ \$guid }}).
Invalid UUID format	Use a standard UUIDv4 format.
Duplicate transactionUUID	Generate a new UUID per upload.
Empty CSV body	Body must be binary CSV; verify Content-Length > 0.
URL spaces	Avoid spaces in the URL between parameters.

9. Postman Quick-Start kit.

Open Postman, then:

- Authorization tab → Bearer Token → paste token.
- Params → transactionUUID = {{\$guid}}.
- Optional Params → scheme = UID | CODE ; strategy = append | update | delete; dry-run = true/false.
- Headers → Content-Type: text/csv (and CamelFileName: <name>.csv).
- Body → binary → select your CSV file (no warning icon).
- Open Postman Console and confirm Content-Length > 0.